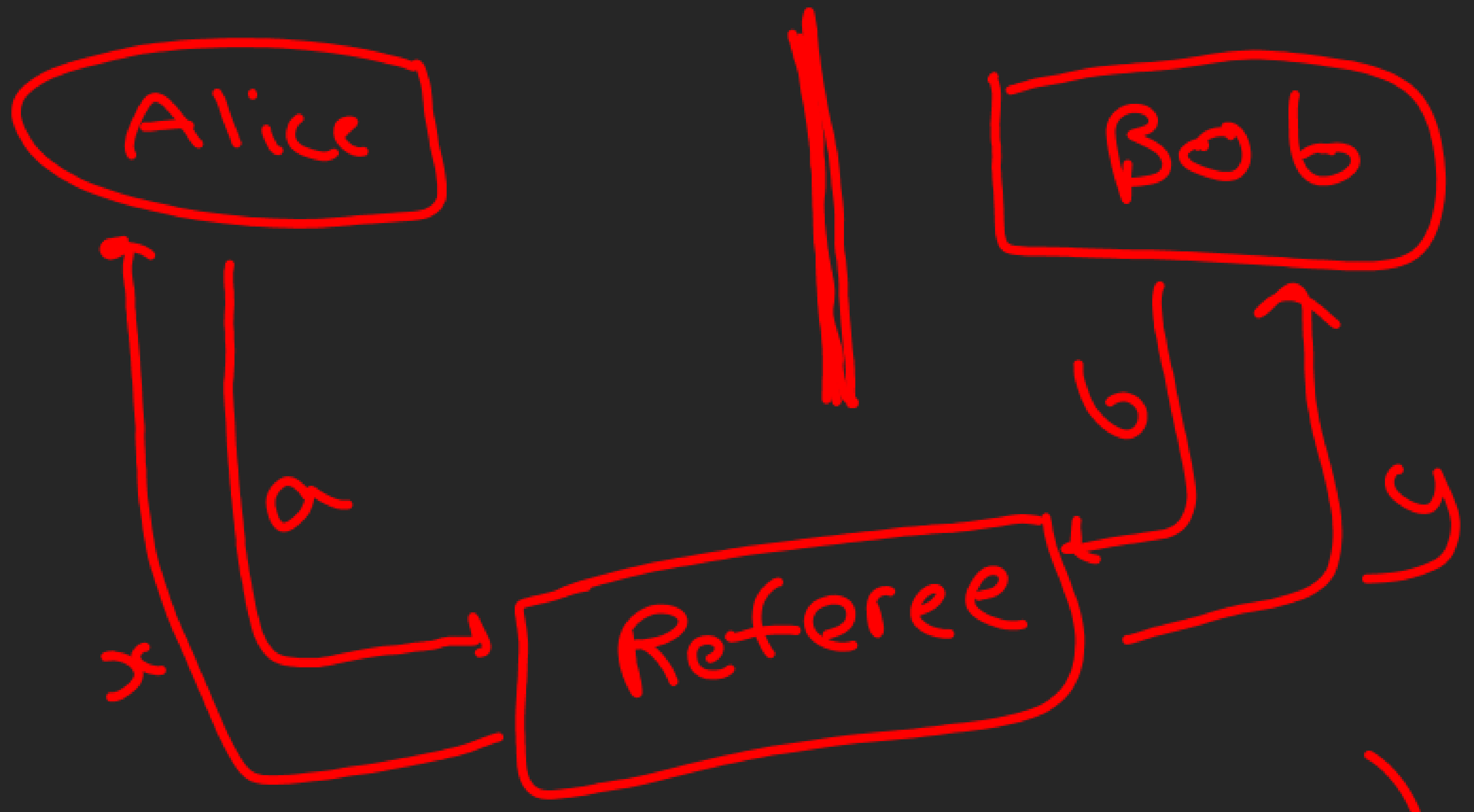
C.H.S.H. Game / Bell Test



→ Alice and Bob have no communication after the game starts

→ Non-local Game

Winning condition → $a \oplus b = x \wedge y$

(x, y)	Winning condition
$(0,0)$	$a = b$
$(0,1)$	$a = b$
$(1,0)$	$a = b$
$(1,1)$	$a \neq b$

→ Deterministic Strategy

$a(0) \oplus b(0) = 0$
$a(0) \oplus b(1) = 0$
$a(1) \oplus b(0) = 0$
$a(1) \oplus b(1) = 1$

→ max win % → 75%

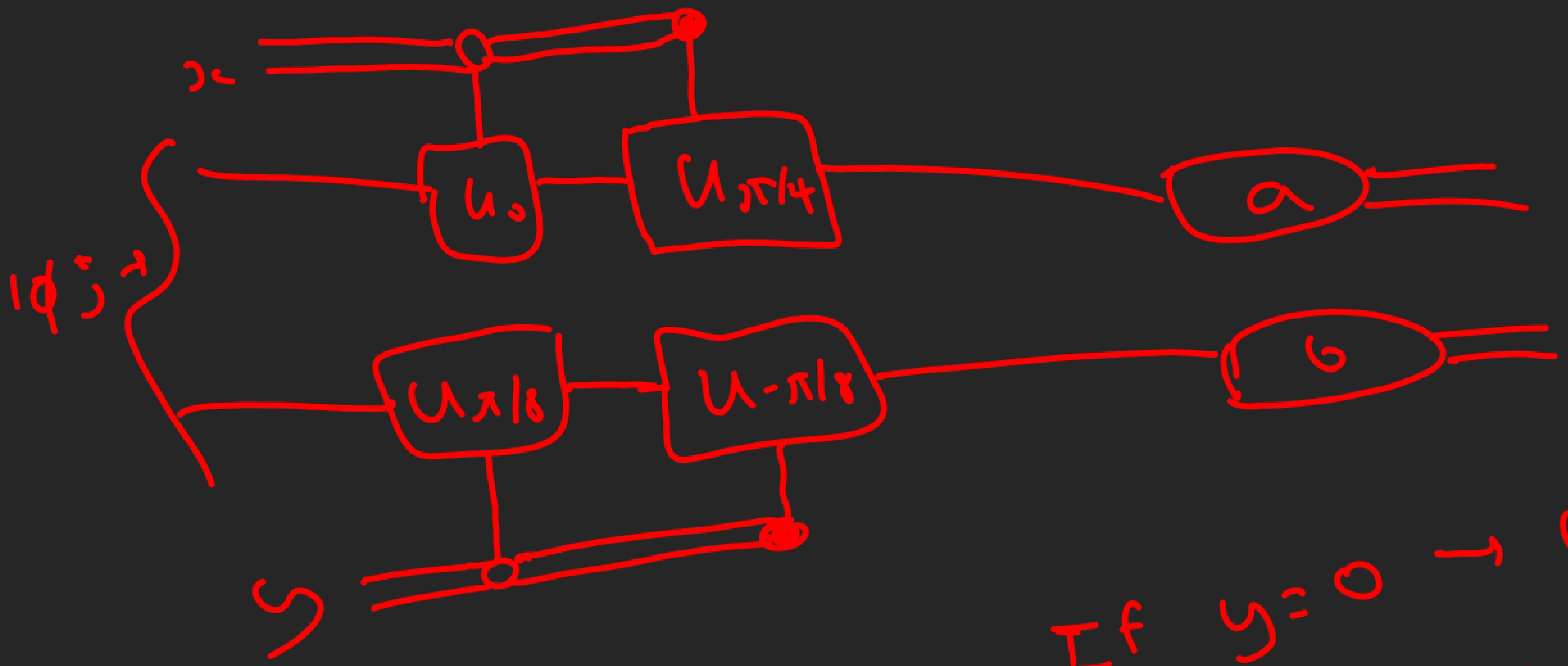
Quantum Strategy

$\langle \psi_\alpha | \psi_\beta \rangle = \cos(\alpha - \beta)$

$|\psi_\theta\rangle = \cos\theta |0\rangle + \sin\theta |1\rangle$

$\langle \psi_\alpha \otimes \psi_\beta | \phi^+ \rangle = \frac{\cos(\alpha-\beta)}{\sqrt{2}} = \frac{\langle \psi_\alpha | \psi_\beta \rangle}{\sqrt{2}}$

$U_\theta : |0\rangle|\psi_\theta\rangle + |1\rangle|\psi_{\theta+\pi/2}\rangle$



If $x=0 \to U_0 = I$

$x=1 \to U_{\pi/4}$

If $y=0 \to U_{\pi/8}$

$y=1 \to U_{-\pi/8}$

For all cases $Pr(\text{win})$ is $\frac{2+\sqrt{2}}{4} \approx 0.85$

$\hookrightarrow \cos^2\left(\frac{\pi}{8}\right)$

Ex. $\rightarrow (x, y) \equiv (0,0)$

$$(U_0 \otimes U_{\pi/8}) |\phi^+\rangle = \frac{\cos\left(\frac{\pi}{8}\right)|00\rangle + \cos\left(\frac{5\pi}{8}\right)|01\rangle + \cos\left(\frac{3\pi}{8}\right)|10\rangle + \cos\left(\frac{\pi}{8}\right)|11\rangle}{\sqrt{2}}$$

$$Pr(a=b) = \cos^2\left(\frac{\pi}{8}\right)$$

$\hookrightarrow$ Win % $\rightarrow$ 85%